Madhan Lankalapalli

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PROFESSIONAL SUMMARY

Digital VLSI Engineer specializing in RTL design, FPGA implementation, and deep learning accelerator development. Delivered a parameterized 4×4 Output-Stationary Systolic Array in Verilog for 2D convolutions and built an end-to-end object-detection pipeline for ADAS on the ZCU104 platform. Proficient in Verilog HDL, High-Level Synthesis, and the full FPGA design flow. Seeking to leverage this experience to contribute to impactful semiconductor and VLSI system projects.

WORK EXPERIENCE

Research Intern – FPGA Communication **Jan 2026 – Present**

SWAN Lab, IIT Kharagpur, West Bengal, India

- Built a real-time FPGA DSP system in Verilog, computing RMS, peak detection, and variance for continuous AQI signal monitoring. Integrated an SPI Master and dual-port circular BRAM into a sequenced time-domain pipeline.
- Modelled and verified a 256-point FFT core with pipelined butterfly units and optimized scaling, achieving exact peak detection and 100% dominant frequency accuracy with less than 0.35% RMS error across diverse signal types.

Research Intern – Object Detection for ADAS **Jun 2025 – Jan 2026**

Model Engineering College, Kochi | Chip to Startup (C2S) Program, MeitY, Govt. of India

- Designed and validated an object detection algorithm for ADAS, moving from Python algorithm testing to Verilog implementation on a ZCU104 FPGA using VITIS AI workflow, which produced a functional hardware prototype.
- Evaluated multiple detection models in Python and selected YOLOv5 for the ZCU104 platform, reducing Verilog conversion effort and shortening prototype development time.

TECHNICAL SKILLS

- **HDL / Programming:** Verilog HDL, C++, Python
- **FPGA Tools & EDA:** Xilinx Vivado, Vivado HLS, Vitis IDE, Vitis AI, MATLAB, MultiSim, LTspice, Arduino IDE
- **VLSI Competencies:** RTL Design, Digital Design, FPGA Design, High-Level Synthesis (HLS), Static Timing Analysis (STA), VLSI Design Flow (RTL-to-GDS), Physical Design
- **Communication Protocols:** UART, I2C, SPI, AXI4-Stream
- **FPGA Platforms:** Nexys A7, ZedBoard, Spartan-6, Tang Nano 20K, ZCU104

EDUCATION

Bachelor of Technology – Electronics and Communications Engineering **2022 – 2026**

Rajiv Gandhi University of Knowledge Technologies (RGUKT AP IIIT), Nuzvid, Andhra Pradesh, India *CGPA: 8.7 / 10*

Relevant Coursework: Digital System Design, VLSI Design Flow, Physical Design, Embedded Systems, Computer Architecture

PROJECTS

4x4 Systolic PE Array for Convolution Operations |GitHub Link ***Sep 2025 – Dec 2025***

Tools: Xilinx Vivado, Python | HDL: Verilog | Protocol: AXI4-Stream | FPGA: Nexys A7

- Executed a parameterized 4×4 Output-Stationary Systolic Array in Verilog for 2D convolution on Artix-7 FPGA, achieving 100% functional coverage against a cycle-accurate Python golden reference with zero numerical error across all 16 PEs.
- Conducted multi-precision hardware analysis across Q8.8, Q12.4, Q12.8, and Q12.12 fixed-point formats; identified Q8.8 as the optimal format at 0.119 W, and Q12.12 pipelining increased DSP utilization by 100% and on-chip power by 32% relative to Q8.8.
- Deployed Laplacian edge detection on a 640×640 KITTI image using the systolic array, decomposed the input into overlapping 6×6 tiles, streamed through Line Buffers for temporal alignment, and reconstructed the final 638×638 feature map from stitched 4×4 output blocks.

Object Detection for ADAS on FPGA (YOLOv5s) ***Jun 2025 – Sep 2025***

Tools: VS Code, Vitis AI, Xilinx Vivado | Language: Python | HDL: Verilog | FPGA: ZCU104

- Trained YOLOv5s on the KITTI autonomous driving dataset, achieving 90.4% Precision, 86.5% Recall, and 90.7% mAP@0.5.
- Quantized the model using Vitis AI, achieving 80% reduction in model size, and generated a deployable XModel for execution on the ZCU104 FPGA.

Pattern Detection Using CNN in Verilog | GitHub Link ***Feb 2025 – Apr 2025***

Tools: Xilinx Vivado | HDL: Verilog

- Implemented a 4-stage pipelined CNN accelerator in Verilog HDL for 8×8 pattern detection within a 128×128 grayscale image, integrating Laplacian edge enhancement, pattern-matching convolution, max pooling, and threshold-based detection as dedicated RTL modules.
- Designed a convolution stage using a parallel multiplier array with a hierarchical adder tree, compressing 36 partial products, achieving a maximum operating frequency of 155.67 MHz, and full image inference in 242 microseconds.

Image Enhancement on OCT Images Using FPGA | GitHub Link ***Nov 2024 – Jan 2025***

Tools: Xilinx Vivado, MATLAB | HDL: Verilog | FPGA: Artix-7

- Applied negative transformation, brightness control, and adaptive thresholding in Verilog for retinal OCT image enhancement on an Artix-7 FPGA.
- Simulated the Sobel edge detection algorithm to identify retinal layer boundaries and flag potential abnormalities.

WORKSHOPS & TRAINING

- **Summer School on Semiconductor Technology & Microfabrication – CeNSE, IISc Bangalore |** Certificate
  Completed hands-on training in wafer processing, photolithography, doping, etching, thin-film deposition, and cleanroom protocols.
- **FPGA Design Trainee Program (C2S, MeitY) – Model Engineering College, Kochi |** Certificate
  Built a Verilog-based FPGA Digital Payment System with core UPI functionalities.
  1st Place in FPGA Design Contest, C2S Program, Model Engineering College, Kochi.
- **High-Level Synthesis for FPGA – Udemy**
  Transformed combinational and sequential circuits in C++ using Vivado HLS, generated RTL IP cores, and verified designs on FPGA hardware.

CERTIFICATIONS

- System Design Through Verilog – NPTEL, IIT Guwahati | Score: 91% | Rank: Top 5% | Certificate
- VLSI Design Flow RTL to GDS – NPTEL, IIIT Delhi | Score: 88% | Rank: Top 5% | Certificate
- VLSI Physical Design – NPTEL, IIT Roorkee | Score: 83% | Rank: Top 5% | Certificate
- Electronic System Design and PCB Design – NPTEL, IIT Madras | Score: 85% | Rank: Top 5% | Certificate
- Verification Methodology – Maven Silicon | Certificate